

	Molecular layer	Estimand tested	Relation to the class definition	Positive rate: disease–state–only vs neither
Bulk RNA–seq (defines the class)	Bulk liver tissue RNA	Disease vs control	Defines the disease–state class	not applicable
Proteomics	Liver protein, DIA–MS	Disease vs control	Cross–assay replication of the same estimand	19.2% vs 2.9%
Single–cell RNA	Single–cell RNA, donor pseudobulk	Disease vs control	Cross–assay replication of the same estimand	5.2% vs 1.6%
Spatial SVG	Spatial transcriptome	Spatial variability; no disease contrast	Distinct property, so an independent test	35.1% vs 12.3%